
APPENDIX D: SITE SURVEY, RF PLANNING & CIVIL/POWER ENGINEERING GUIDELINES

5G FWA PROGRAM (5,000 SITES NATIONWIDE) Tender Reference: RFP-TELTECH-FWA-2026-001 Project: Nationwide 5G FWA Project (Band n50, 60 MHz TDD)

Operator: Telecommunications Technology Solutions Incorporated (TelTech)

Author: OLIVER TUNGOL --- ## 1. ENGINEERING PRINCIPLES & SITING

ARCHITECTURE ### 1.1 Siting Philosophy

The primary engineering objective of the 5G FWA rollout is to maximize household population coverage within high-density suburban and rural barangay settlement clusters while minimizing civil capital expenditures and permitting lead times.

- **TowerCo Co-Location First (Option A — Baseline):** The Operator leverages an audited master database of **13,746 operational towers**. For every candidate barangay, field teams must prioritize existing towers within $\leq 2.0\text{km}$ (Classes T1, T2, and T3).
- **Greenfield Monopole (Class T4):** Restricted strictly to candidate barangays located $> 2.0\text{km}$ from any verified operational tower structure ($\sim 9.3\%$ of sites nationwide; 103 sites in Phase 1). --- ## 2. DETAILED TECHNICAL SITE SURVEY REPORT (TSSR) REQUIREMENTS For every nominal site candidate, the Contractor shall complete and submit a comprehensive TSSR conforming to the following structure: ### 2.1 General Site Identification & Geography
 - Site Name, Candidate ID, Official PSGC Code, Region, Province, Municipality, and Barangay.
 - Precision WGS84 GPS Coordinates (Latitude, Longitude recorded with ≥ 6 decimal places via dual-frequency GPS receiver).
 - Ground Elevation Above Mean Sea Level (AMSL in meters).
 - Access Road Assessment: Road surface type (concrete, asphalt, macadam, dirt trail), bridge weight limits, and 4x4 hauling accessibility during monsoon seasons. ### 2.2 Tower Structure & Loading Audit (Co-Location Sites)
 - Operating TowerCo Entity (PTCI/MIDC, FTAP Globe, EdgePoint, FTAP Smart, LDIC, FTAP BTS).
 - Tower Type (Self-Supporting 3-Leg/4-Leg Tower, Monopole, Guyed Mast, Rooftop Tower).
 - Tower Height Above Ground Level (AGL in meters).
 - Structural Space Audit: Proposed mounting height for 3x Band n50 antennas (recommended centerline: 30m to 45m AGL).
 - Structural Loading Verification: Confirmation that existing tower members and foundation can accommodate additional 3x 8T8R RRUs/AAUs ($< 45\text{kg}$ each) and wind load profile of 250 km/h.

- Panoramic Drone Photographic Survey: High-resolution 360° continuous panoramic video and 12-cardinal-point still photos at proposed antenna centerline height to verify optical line-of-sight across populated settlements. ### 2.3 Power Infrastructure & Utility Audit
- Existing Commercial AC Utility Power: Electric Cooperative / Distribution Utility name (e.g., MERALCO, PELCO, BENEKO, CAPELCO, ILPI).
- Service Entrance Capacity: Transformer rating (kVA), service voltage (230V / 400V, 60 Hz), breaker rating, and phase balance.
- TowerCo DC Power Availability: Sourcing from TowerCo DC busbar (-48V DC) vs. installing Operator-dedicated rectifier subrack.
- Sub-metering feasibility and cable tray routing for power cables.
- Grounding System Audit: Measurement of existing earth ground resistance using a 3-point fall-of-potential earth tester. Earth resistance must be <= 5.0 Ohms. ### 2.4 Backhaul Transmission Path Survey
- Audited Codex Optical Screening Match: Distance to nearest confirmed node candidate (Class A <= 0.5km, Class B <= 1.0km) or line-tap closure (Class C <= 0.5km).
- Cable Entry Facility: Manhole, handhole, or aerial pole lead-in routing into tower shelter/compound.
- For Remote Class E/F Sites: Satellite antenna mounting location (unobstructed sky view with 0° to 80° northern elevation arc for Starlink constellation tracking). --- ## 3. RF PROPAGATION & CELL DIMENSIONING GUIDELINES ### 3.1 Link Budget Parameters (Band n50, 1.5 GHz) | Parameter | Unit | Value (Downlink) | Value (Uplink) | Technical Notes |

Operating Frequency	MHz	1475.0 (Center)	1475.0 (Center)	Band n50 TDD
Channel Bandwidth	MHz	60.0	60.0	162 Resource Blocks (SCS 30 kHz)
BTS Maximum Conducted Power	dBm / W	55.0 dBm (320 W)	N/A	8 x 40W RRU configuration
BTS Power per Resource Block	dBm	32.9 dBm	N/A	Uniform power distribution
BTS Antenna Gain	dBi	17.5 dBi	17.5 dBi	8-port cross-pol macro antenna
BTS Cable / Jumper Loss	dB	1.0 dB	1.0 dB	Short optical/RF jumper interface
CPE Maximum Transmit Power	dBm / W	N/A	23.0 dBm (200 mW)	Power Class 3 UE (or Class 2 26 dBm)
CPE Antenna Gain	dBi	9.0 dBi	9.0 dBi	High-gain directional indoor/outdoor
CPE Body / Indoor Penetration Loss	dB	8.0 dB	8.0 dB	Concrete hollow block & glass attenuation
Foliage / Clutter Margin	dB	6.0 dB	6.0 dB	Typical rural tropical tree canopy
Shadowing Margin (Log-Normal)	dB	7.0 dB	7.0 dB	90% edge cell coverage reliability
Thermal Noise Density	dBm/Hz	-174.0	-174.0	kT @ 290K
Receiver Noise Figure	dB	6.0 dB (CPE)	3.5 dB (gNodeB)	Active RF front-end
Target SINR for Min Service	dB	-1.0 dB	-2.0 dB	QPSK rate 1/3 (Cell Edge 25 Mbps DL)
Maximum Allowable Path Loss (MAPL)	dB	144.5 dB	139.8 dB	Uplink is coverage limiting

| **Calculated Reliable Cell Radius**| km | **3.5 km** | **3.2 km** | **Standard suburban/rural clutter** | ### 3.2 Nominal Sector Azimuth & Tilt Strategy

- **Sector Count:** 3 sectors per macro site with 120° nominal separation (0°, 120°, 240°). Azimuths must be tailored during TSSR to track actual barangay population concentrations along transport corridors and coastal plains.
- **Electrical Remote Downtilt (eRET):** Antennas must feature AISG 2.0 remote tilt adjustable between 2° and 10° from the central OSS. Mechanical tilt shall only be applied when required to clear close-in obstructions (>15°). - -- ## 4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD SITES) For the ~10% of sites requiring self-constructed greenfield towers, the Contractor shall adhere to the following civil standards:

1. **Monopole Tower Specification:** 30-meter or 40-meter polygonal tubular steel monopole constructed in 3 to 4 slip-joint sections. Structural steel: ASTM A572 Grade 50 (or equivalent high-strength structural grade) hot-dip galvanized per ASTM A123 (>= 85microns zinc coating). Design Wind Speed: Minimum **250 km/h 3-second**

gust in accordance with the National Structural Code of the Philippines (NSCP 2015 / TIA-222-G/H). Top antenna mounting platform capable of supporting 6x antennas/RRUs and microwave dish mountings.

2. **Compound Civil Works:** *Compound Footprint: Standard 8.0m x 8.0m (64 m²) gravel-topped compound. Perimeter Fencing: 2.4-meter high cyclone wire mesh fence topped with 3 strands of galvanized concertina razor wire. * Equipment Pad: Reinforced concrete equipment plinth (Class A concrete, 28-day compressive strength $\geq$ 21MPa / 3,000psi) for outdoor cabinets and generator set.*
3. **Lightning Protection & Grounding Grid:** *Air terminal (lightning rod) installed at monopole top extending $\geq$ 2.0m above uppermost antenna. Down-conductor: 70mm² bare stranded copper cable routed externally and clamped every 1.0m to tower shaft. * Earth ground ring: Continuous buried bare copper loop around tower base and compound with copper-clad steel ground rods (19mm dia x 3.0m length) exothermically welded (Cadweld) at all junctions, achieving $\leq$ 5.0 Ohms. --- ## 5. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA Before any site is granted Commercial On-Air Acceptance, the Contractor must complete and submit a signed SSV Report verifying:*
4. **Stationary Functional Tests:** *Sector Verification: Physical azimuth and tilt match TSSR approval within $\pm$ 2°. RF Transmit Power: Conducted power on all 8 transmit paths within $\pm$ 0.5 dB of nominal. VSWR / Return Loss: Return loss $\geq$ 15 dB across all antenna feeder lines. Max Throughput Test: Single CPE situated in line-of-sight achieving $\geq$ 450 Mbps DL and $\geq$ 60 Mbps UL.*
5. **Short Drive Test & Static Multi-Point Test:** *5 stationary measurement points per sector (Near: 500m, Mid: 1.5km, Far/Edge: 3.0km). RSRP $\geq$ -95 dBm across $\geq$ 90% of surveyed barangay inhabited zone. Ping RTT $\leq$ 20 ms to local UPF IP address; zero packet loss over 1,000 continuous pings. Inter-sector handover execution without drop during transition. ---*

(End of Appendix D — Site Survey, RF Planning & Civil Engineering Guidelines)